Impact of Federated Learning On Smart Buildings

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***Abstract*—Advances in Internet of Things open up many new possibilities in the design of smart-buildings. Thanks to the wide deployment and the growing computing power of connected devices, the researchers can now consider in-situ with distributed learning processes where on connected devices, local learners collect data to improve the building management system knowledge. In-situ and distributed learning is a convincing direction for smart-buildings since they can better take into account users contexts, privacy concerns or the minimization of energy consumption. However, this direction introduces new challenges such as that of formulating distributed learning processes in which data are collected locally, aggregated to generate models which themselves serve to define a global intelligence. This paper considers the formulation of such a distributed learning process, Federated personalization : a distributed learning strategy where individual learners collaborate in a knowledge sharing round to benefit from shared generalization, yet retain local specificity. This research work proposes a federation framework to orchestrate a group of autonomous learners naturally distributed across smart buildings. The results show that leveraging inter learner knowledge transfer makes it possible to achieve personalized lower generalization loss for regression problems like forecasting time series data and image classification. The prototype is currently deployed at Qarnot, a French computing heater company to provide secure and sustainable autonomic ambient intelligence via edge devices.**

***Index Terms*—Smart Buildings, Edge Computing, Federated Learning, Internet Of Things, Ambient Intelligence, Multi sensor Networks**

I. Introduction

The evolution of smart buildings [1] is powered by technological growth for seamless control and optimal comfort of indoor/outdoor operations. One of the pillars driving technical growth is the surging popularity of using IoT (Internet of Things) products due to improved quality of service, inexpensive assemblage, ease of installation, Do-It-Yourself(DIY) tutorials, etc. A recent European Union report [2] described a wide range of challenges for a smart building integration into a smart city. Post COVID, with companies offering "work from home" schemes, the rise in home base automation has gone up, making smart building industry ever more interesting to work on. A chunk of smart building challenges is addressed by artificial intelligence, which is able to deliver reliable results for example in energy monitoring, appliance scheduling, ventilation modulation, comfort regulation etc. Historically, machine learning for smart-building consisted mainly in centralized ex-situ processes. Here, data collected from predefined scenarios and benchmark were used to define the decision models of building management systems. With ex-situ learning, the specific habits of the inhabitants living in a house were not necessarily taken into account: the data might not be representative.

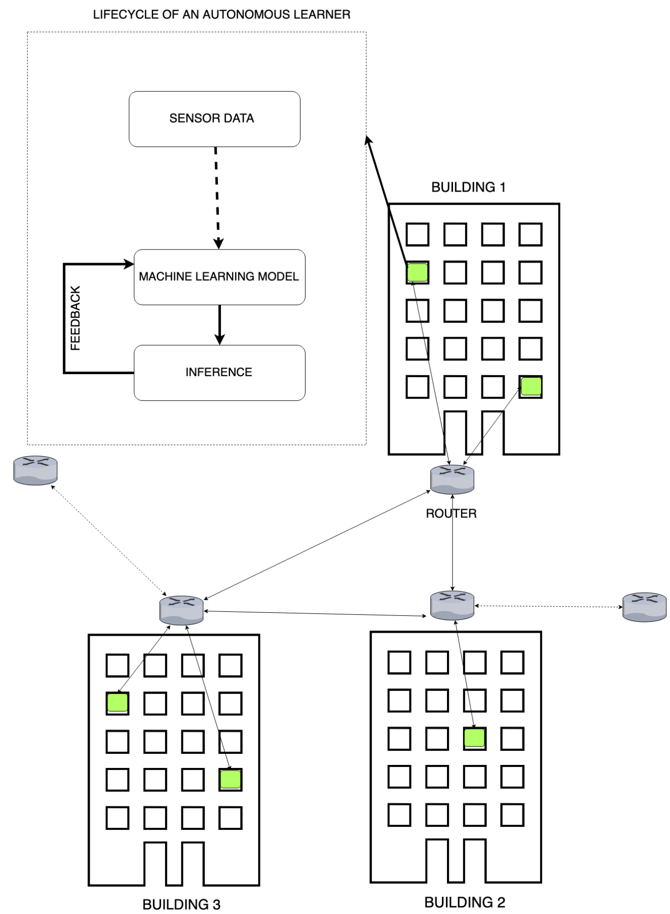

Fig. 1. Autonomous learning agents [in green] placed in rooms across buildings connected within a smart city.

Thanks to advances made in IoT, in-situ with distributed learning is nowadays a more promising decision. The idea is to consider learning processes, in which the computing and sensory power of connected devices serve to collect data and generate local models then combined to develop collective intelligence. In-situ learning can lead to more qualitative models as it is based on contextual data. The ability to learn locally is also interesting for privacy and network consumption

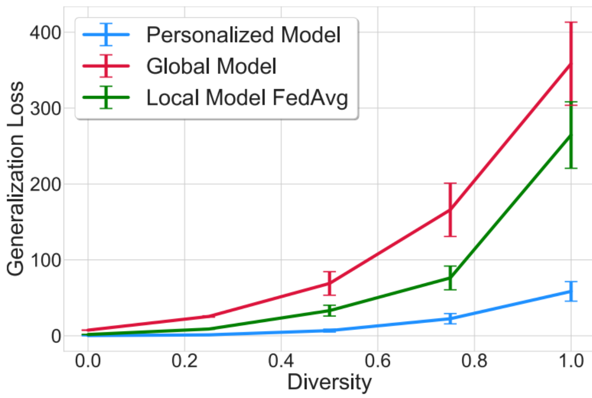

Fig. 2. Improved Model Performance due to Federated Averaging [3]

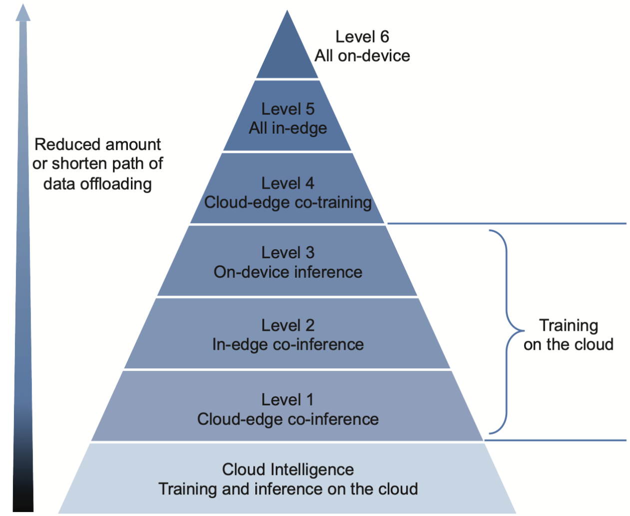

Fig. 3. A 6 level rating for edge intelligence [9]

since we can envision to not transfer raw data but compressed ones. However, there are several challenges to consider: how do we collect contextual data? how do we transform them into knowledge? how do we measure the quality of machine learning models? how do we define the global error? These are some questions whose answer is not easy when considering contextual data processed by a distributed process.

In this paper, we consider the implementation of in-situ and distributed learning process throughout Federated personalization: a distributed learning strategy where individual learners collaborate in a knowledge sharing round to benefit from shared generalization, yet retain local specificity. Our proposed federation framework [Fig 1] performs decentralized orchestration amongst autonomous learning agents to solve regression or classification tasks. The work is a proof of concept with real life deployment that shows promising incentives for knowledge sharing to achieve lower generalization errors on machine learning tasks for a federated learner compared to a learning only from local data strategies.

II. Related Works

The context of this work addresses artificially intelligent models auto-updating via federated learning processes running on edge computing devices deployed in smart buildings.

A. Federated Learning

Data generation across a vast number of devices acts as a bottleneck to train a machine learning model with a centralized data. Federated learning [4] aims to mitigate training complexity by splitting the learning process into rounds to incorporate the knowledge stemming from evolving data. These methods perform well if local distributions are similar but the authors [5] explicate that a global model via Fed-Avg may diverge due to high sensitivity on hyper parameter tuning and not generalize well [6] on local data due to highly non IID data. The increase in data heterogeneity [7] with growing number of clients aggravates the generalization error. [3] in [3] introduce a bi-level optimization method, which to reduce local generalization error as shown in Fig. 2. The process regarding federated learning is usually a synchronous federation process [8] where

1) Mediator selects a subset of local learners each of which downloads a version of the global model.
2) Each selected local learner computes an updated model on their local data.
3) The updated model weights are sent to the server or global node from the selected clients.
4) The server aggregates these models (typically by Fed-averaging) [4] to construct an improved global model.

Our work provides a new alternative to initiate and orchestrate a federation process without a role of a mediator enabling peer to peer learning that can be extended to support federated learning within groups of autonomous learners. The framework offers the flexibility to perform federated learning on an arbitrary network of spatio-temporal data generating sources.

B. Edge Intelligence

The IoT paradigm caters to low powered computing resources that are distributed across the edge. [10] summarises existing literature on Edge Intelligence (EI) as a confluence between edge computing and Artificial Intelligence, born out of the philosophy to process sensor data in-situ, with low quantity and high quality data transmission, and low computing power requirement. Around 850 ZB of data is expected to be generated by mobile and IoT devices by 2022 [11] which raises significant operating concern for cloud volumes. The paper [9] introduces a 6 level pyramid shown in Fig. 3 to measure edge intelligence based on execution of training and inference in cloud, edge or in-situ on device. The pyramid base (Level 1) is least edge intelligent due to training and inference on-cloud while the top most layer (Level 6) corresponds to highest edge intelligence with complete on device execution. A recent area of interest in EI [12] [13] have been in porting deep learning modules to edge [14], specifically to reduce computation and transmission of millions of training parameters over network. The work proposed is Level 6 edge intelligent framework that can also pre-estimate retentive storage capacity needed to save model parameters, approximate data transfer in a federation round and provides deployment flexibility via adjustable parameters to fit according to device storage constraints.

C. Smart Building

Remote controlled operations of sensors or actuators in a smart building is an application of edge intelligence. [15] propose a definition of a smart building by quantitatively categorising smart buildings based on a metric expressed as the ratio between the number of controllable and observable building related parameters [16]. [17] introduces a middleware for context-aware ubiquitous computing and points to the need to balance between transparency and context-awareness and the requirement for a certain degree of autonomy. Considerable amount of research has been done on reducing energy consumption [18], [19], [20], maintaining indoor air quality [21]. A building is usually a closed context of space and time where one can often find repetitive activities like cooking, eating etc. Forecasting sensor values is important to maintain reactive Heating, Ventilation and Air Conditioning (HVAC) systems, detect anomalies. [22], identify activities [23], [24]. [25] extend the scope of the problem to update/evolve a context-aware model for classifying situations based on tagged multi-modal data and feedback. The work of [26] shows a potential in energy saving by proposing deep learning based forecasting models. These ex-situ models are generally trained on labelled and cleaned data that is often scarce/ infeasible to generate at the edge. Our contribution mitigates the problem by presenting a generic framework of autonomous learners that can auto-update periodically with limited data and is robust to issues of distribution diversity. Furthermore a learner performing in-situ data predictions can benefit from knowledge sharing with other learners equipped with similar spatio-temporal data sensors.

III. FEDERATION FRAMEWORK

We present a framework that enables autonomous learning processes running on edge to collaboratively learn from a group of peers by broadcasting a configuration call to federate, followed by a round of local knowledge upload to an edge storage with limited capacity. We breakdown the framework into the following components that can be combined together to support a wide range of distributed learning algorithms.

A. Autonomous Learner

Autonomous Learner (AL) is a set of processes executing on an edge device, responsible for training and drawing regression or classification based inference on sensor data generated at site in real time. It can create a deep learning model h specific to a task T and is responsible to update the model over

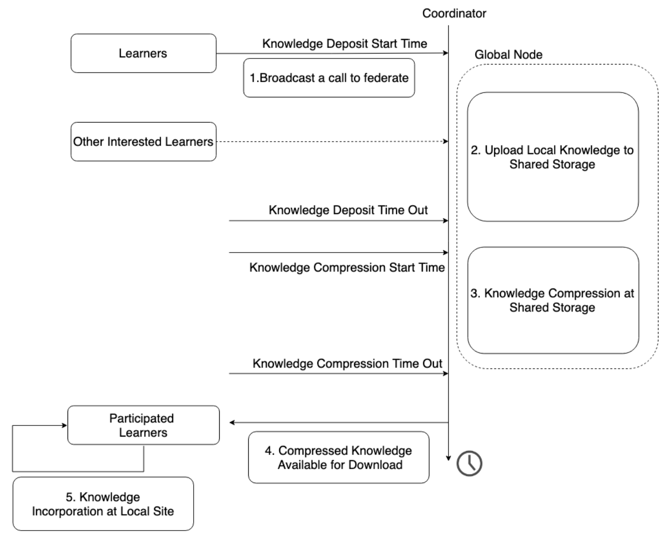

Fig. 4. Chronologically executed steps for a synchronous federation round.

time to minimize generalization loss. A **Local Data Buffer** stores a limited amount of historical sensor data mainly used for retraining and validating model updates and as a backup of latest model parameters. Typically a buffer can hold past 1-2 weeks of data and is refreshed on arrival of incoming data. Sensor data collected from i^{th} Autonomous Learner from time t_1 to t_2 is represented as $X^i_{t_1:t_2} \in R^{dXk}$ with $d = \frac{t_2-t_1}{\Delta resolution}$ rows and k sensors. Setting $\Delta resolution$ to 5 minutes yields 288 samples a day for example. We empower the learner to communicate with other learners for knowledge up-gradation but is prohibited to share raw sensor data due to privacy and storage issues. The optimization solved $h_i^t \leftarrow \arg\min(L(h_i^{t-1}, X^i), L(\hat{h}^t, X^i))$ where $\hat{h}$ is model mixed with external knowledge. $\hat{h}^t$ is constructed as $\hat{h}^t \leftarrow \beta_D h_i^{t-1} + (1-\beta_D) h_G$ where β_D is a tune-able parameter that measure the distrust factor with the global model and h_G is the global weight average. Setting $\beta_D = 1$ will ensure a local model is not affected by global model updates.

B. Global Node

A global node acts an headless server who orchestrates the process of federation. It can initiate, monitor and plan the stages of learning. It can create the initial weights for learning, and as the rounds progress, it receives and stores the updated parameters. Figure 4 shows the stages of federation in a synchronous round where a federation call starts by broadcasting the Knowledge Deposit Start Time indicates for selected learners to upload their knowledge representation to a on shared storage like edge disk or a cloud hosted storage like an Amazon S3, Google Cloud Compute Bucket with read/write permissions for collaborating AL. After timeout, it will average the weights accumulated in a round in order to provide new global parameters. The global parameters are sent to any component that sends a request message. This is the consolidation stage where an aggregate function running

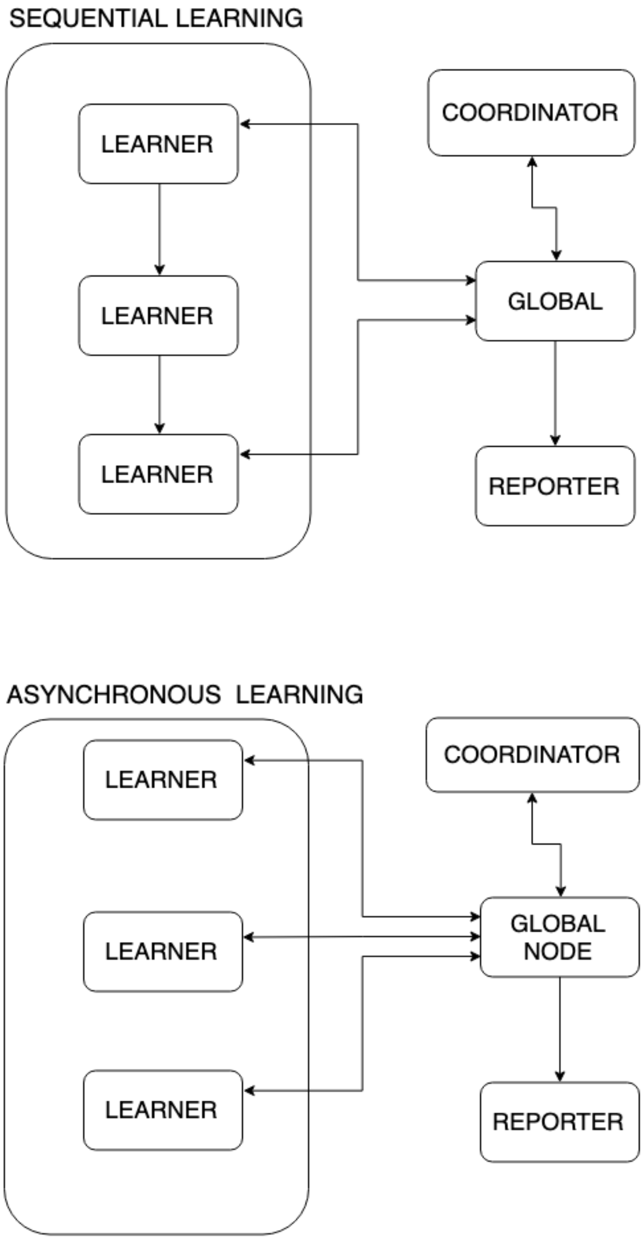

Fig. 5. Framework Configuration for sequential and asynchronous task execution modes.

on shared storage compressing the uploaded weights with FedAvg [4]. Formally, a collaborative federation is represented by (xE, yR) where x denotes epoch (E) or frequency of using a data point locally during a training update and y indicates the number of rounds (R) federation takes place. For example, a model restricted to train only on data locally available for 20 epochs is given by (20E, 0R). A cycle of federation is repetition of federation rounds, each of which can broken down into logical 5 steps as shown in Fig. 4.

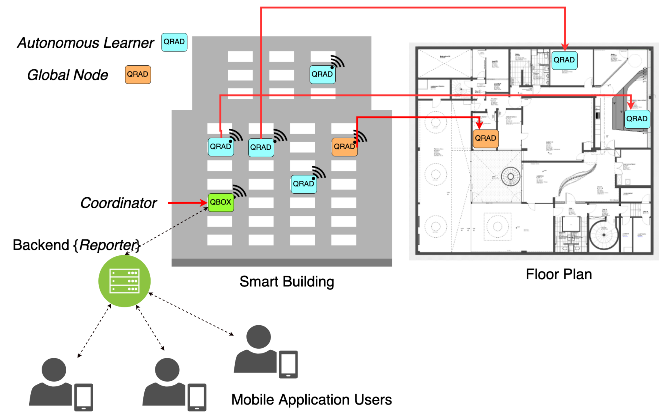

Fig. 6. Framework Components running on the Qarnot Infrastructure

C. Coordinator

It is responsible to send or receive messages on the message bus to control the learning process. The coordinator can be a stand alone component, or can live in a process with learner or global nodes. Examples of coordination messages could include a call to federate, or a request to complete a learning round. Compressed Knowledge is made available for download to autonomous learners by broadcasting the termination of a successful round.

D. Reporter

It listens for progress reported by components in the learning process and saves this diagnostic information to a data source in order to later generate reports. The component acts as the liaison between edge and cloud services by delivering actionable data on state of operation to user interfaces via mobile apps and building management software.

IV. Experiments

We have setup this framework on Qarnot's edge computing heater (Qrad) that are placed on a floor of a smart building as shown in Fig. 6. Each Qrad is equipped to measure temperature (in Celcius), relative humidity, noise levels (in decibels), power (in Watts), luminous intensity (in lux) and indoor air quality (in ppm). The test-bed consists of 1.4 Million Qrad sensor data readings, sampled at every 5 minutes leading to 288 samples a day, totalling to 208 k samples in 8 months spread across 3 rooms namely kitchen, meeting room and open office.

A. Forecasting Model Performance

The first experiment is to observe the impact of retraining on sequence to sequence predictions of sensor value forecasting on a future period. For the forecasting task, we implement a deep learning model that has a look back of 36 timestamps or 3 hours, feature size of 6, 1 hidden layer, with a fully connected last layer of $12 \times 6 = 72$ outputs or 1 hour prediction. One variant of the model has 50 Long Short Term Memory (LSTM) units with 11200 hyper-parameters (40-45 kB). The other

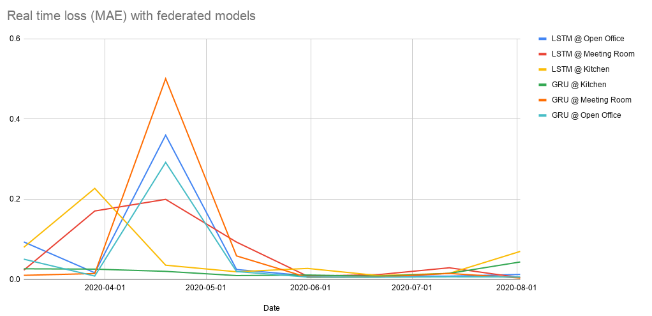

Fig. 7. Mean Average Error of 6 auto-updating models distributed over 3 spaces with 2 models per room over 8 months.

Update Date	LSTM @ Office	GRU @ Office	LSTM @ Kitchen	GRU @ Kitchen
2020-03-08	0.093	0.050	0.080	0.026
2020-03-29	0.016	0.008	0.227	0.025
2020-04-19	0.359	0.292	0.035	0.019
2020-05-10	0.024	0.019	0.018	0.009
2020-05-31	0.008	0.005	0.027	0.010
2020-06-21	0.008	0.005	0.009	0.007
2020-07-12	0.007	0.006	0.014	0.014
2020-08-02	0.012	0.005	0.069	0.043

TABLE I
GENERALIZATION ERROR OF LSTM AND GRU MODELS EXECUTING IN A KITCHEN AND OPEN OFFICE AREA.

variant has 32 Gated Rectified Units (GRU) with 4000 hyper-parameters (<10 kB). Adam optimizer and mean absolute loss functions are used to train and evaluate both the deep-learning models. Table I illustrate the generalization loss for both forecasting models applied to 3 rooms periodically updated after every 3 weeks. In Fig. 7 we observe loss peaks at March-April which coincides with the distinct spatio-temporal pattern of reduced office activity during the lockdown in France. We split 8 months of data to perform a 4 fold cross-validation, with average residual error between 7-12 % higher than retrained forecasting models.

B. Impact of Federation Configuration

Now we illustrate the impact of selective or pairwise federation on loss values of individual forecasting models placed at an open office, a meeting room and a kitchen as shown in Fig. 8. We observe that the meeting room model records 7.6 % lower generalization error post federating with a kitchen (in yellow) than a distant open office (in red). Compared to a non-federated version, the federated open office model learns better from the kitchen (12 % improvement) rather than the meeting room (2.5 % improvement). The intuitive validation lies in people gathering naturally at both the areas during breaks or an informal meeting. Both the observations indicate similar spatio-semantic ontology may play a key factor in the process of federation.

We validate the impact of the federation by comparing against a setting without knowledge transfer. Hence the baseline model follows zero round (0R) strategy and is illustrated with 2 federation configurations out of infinite combinations of

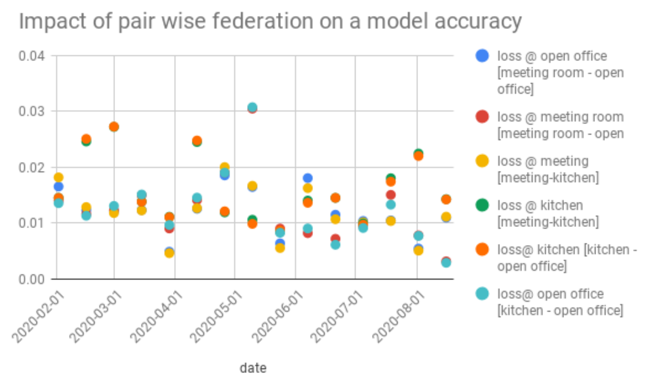

Fig. 8. Pairwise Federated Loss curves of unit layered LSTM models placed at 3 rooms.

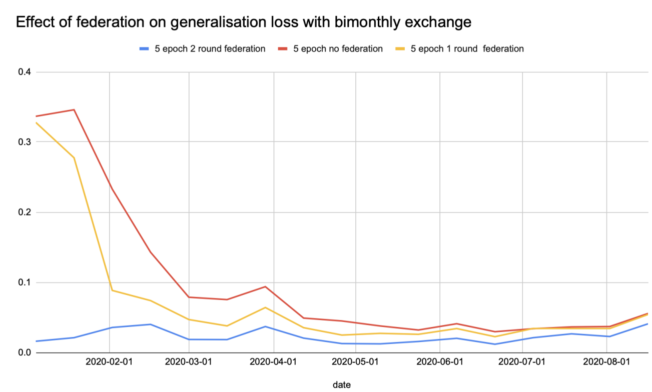

Fig. 9. Performance gain due to federation

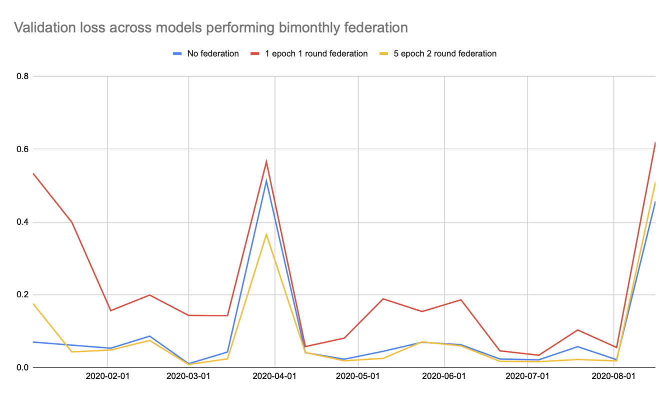

Fig. 10. Under-fit and over-fit federated models vs baseline non-federated model performance.

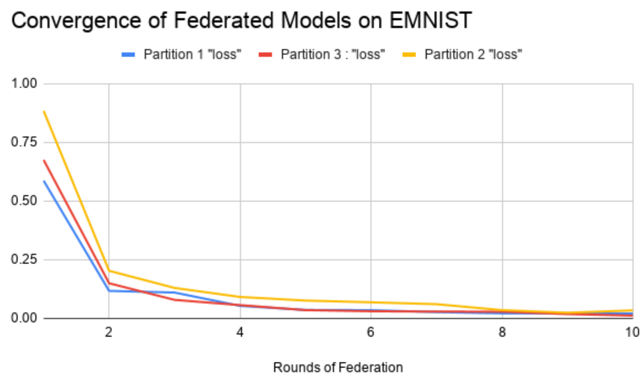

Fig. 11. Convergence of federated models on EMNIST data-set.

Round	Partition 1	Partition 2	Partition 3
1	0.587	0.885	0.675
2	0.118	0.203	0.150
3	0.110	0.130	0.079
4	0.054	0.092	0.057
5	0.036	0.076	0.036
6	0.035	0.069	0.031
7	0.027	0.061	0.029
8	0.022	0.036	0.029
9	0.022	0.024	0.019
10	0.021	0.034	0.012

TABLE II
AVERAGE MEAN ABSOLUTE ERROR OF FEDERATED MODELS ON 3 PARTITIONS OF EMNIST DATASET.

(x epochs,y rounds). In Fig. 9 we plot the average generalization error of three collaborating learners after every bimonthly update across 8 months. We observe a (5E,2R) federation configuration experiences lower generalization loss over a non federated (5E,0R) and less federated (5E,1R) solution. It is evident there is no one fit for all but certain combinations can lead to an under-fit model (5E,1R) performing worse than a non-federated local learning strategy (5E,0R) as shown in Fig. 10.

C. Federated Computer Vision Tasks

The third scenario evaluates the power of federation in computer vision tasks like facial, handwriting character recognition etc. In a distributed setting, each edge device has an access to a partition of images and knowledge sharing is expected to increase the collective cognition of the system. The task is to recognize hand written digits from a bifurcated EMNIST data-set with each partition containing 3000 training and 1000 testing samples. Fig 11 illustrates convergence of 3 deep learning models (2 layers CNN, with 32 hidden units) each less than 40k parameters within 6 rounds with 98 % accuracy with a (2E, 10 R) strategy, the loss values are recorded in table II.

V. CONCLUSION

Qarnot has over 2000 units of smart computing heaters deployed in over 500+ homes and office spaces. In an office setting, the heaters are usually connected via the local area network and this enables the heaters to communicate with one another. The novelty of the framework is in orchestration of auto-updating deep learning models in real time. The work presented shows promising results about the utility of federated personalization in the domain of smart building. It also contributes to leveraging controllable federated techniques across connected objects in a smart city to provide autonomous edge intelligence.

Diversification in data sources is a bottleneck to federated learning due to the trade-off between quantity of learners/models and quality of data shared. A promising scope for further studies is modelling reinforcement strategies for autonomous learners to select collaborative peers or cluster representatives. Some key investigation clues are figuring out under what conditions does update of an autonomous leaner converge to a local minimum, estimating convergence rates and generalization bounds.

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